From Agriculture to Services: Development without Industrialization

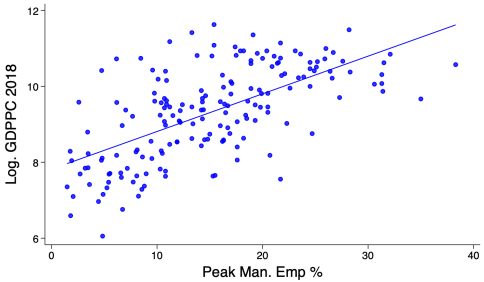
Tancredi Rapone[1,2]

STEG-BSE November 2026

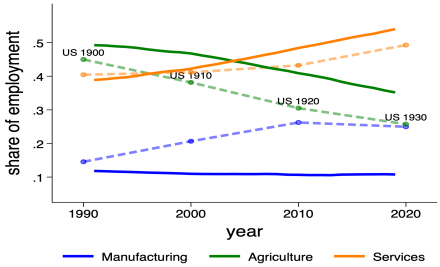
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Manufacturing Matters — But Many Countries Skip It



(a) Manufacturing peaks and income gaps



(b) Different paths into services

Notes: In panel (b), dashed lines show the United States starting in 1900; solid lines show the unweighted average of low- and middle-income countries starting in 1990.

Sources: ILOSTAT and Gapminder.

▸ VA

This Paper: Structural Change and Development

The traditional view of development

Kuznets (1966); Chenery and Syrquin (1975); Herrendorf, Rogerson and Valentinyi (2014)



A new development path

Rodrik (2016)



Does industrialization change the service economy that follows?

Mechanism: Manufacturing Shapes the Service Economy

Large-scale manufacturing

Coordination problems

hire →

High-skill producer services

Managers, engineers...
reusable methods.

diffuse →

Capabilities for services

Service firms adapt

Historical examples of reusable methods

DuPont

Return-on-investment accounting

General Motors

Multidivisional organization

Main idea: Manufacturing changes how production is organized.

Historical Example: Managerial Accounting

1914
DuPont
Manufacturing origin



ROI

1922–24
McKinsey
Codification



1935
Marshall Field's
Transfer to services



McKinsey becomes CEO

What I Do and What I Find

Document	**Cross-country stylized facts** Larger manufacturing peaks are associated with more high-skill producer services (HSPS) and faster growth after manufacturing declines.
Identify	**U.S. counties: shift-share evidence** Exposure to national growth in high-throughput manufacturing predicts larger industrial peaks and greater HSPS specialization in 2000.
Quantify	**Structural Change Model: Endogenous Skills and Growth** Scale creates demand for skills, skills lower cost of education and create growth. **98%** of the HSPS **40%** of the cross-country growth gradients.

Policy: Scale reforms strengthen the lasting effects of support for skills and organizational investment.

Literature

Literature and Contribution

- **Structural transformation and service-led development**

  Herrendorf et al. (2014); Rodrik (2016); Buera and Kaboski (2012); Buera et al. (2022); Fan et al. (2023); Huneeus and Rogerson (2024); Fujiwara and Matsuyama (2024); Sposi et al. (2026); Peters et al. (2026).

  ⇒ I study how manufacturing history shapes service composition and subsequent growth.

- **Firm scale, skills and organizational capital**

  Chandler (1977, 1990); Atkeson and Kehoe (2005); Cox (2025); Engbom et al. (2025); Gottlieb et al. (2026)

  ⇒ I link firm scale to skill demand and the creation of reusable organizational procedures.

- **Industrial policy: big push, increasing returns and persistence**

  Rosenstein-Rodan (1943); Hirschman (1958); Murphy et al. (1989); Juhász (2018); Incoronato and Lattanzio (2025); Choi and Levchenko (2025).

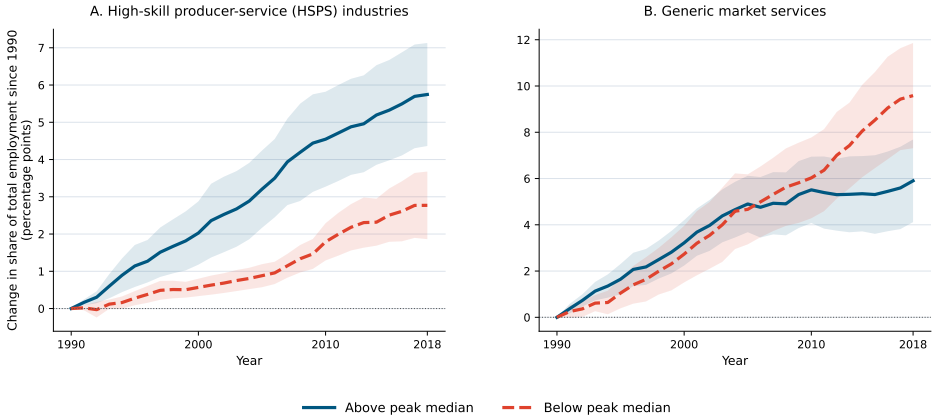
  ⇒ I show how industrial policy can have lasting effects **beyond manufacturing**.

Roadmap

1. **Cross-country stylized facts**
2. U.S. county evidence
3. Model: organization and growth
4. Quantitative analysis
5. Policy implications

Cross-Country Facts

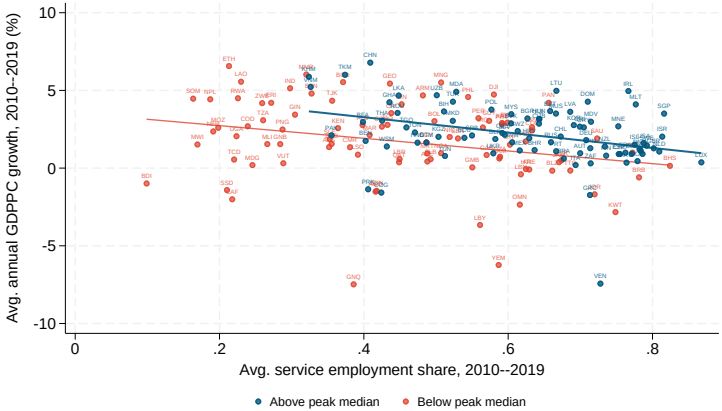
Stylized Fact 1: Manufacturing and Service Composition



51 economies; median manufacturing-peak split. Employment shares rebased to 1990; unweighted means ±1.96 SE. HSPS: business services, finance and real estate. Source: ETD and GGDC.

▸ Regression table ▸ Decomposition

Stylized Fact 2: Manufacturing and Service-Era Growth



2010–2019 averages; median manufacturing-peak split. Source: GGDC, ILOSTAT and World Bank/Gapminder.

▸ Regression table

U.S. County Evidence

Which Manufacturing Industries Require Production at Scale?

High-throughput manufacturing

Iron and steel; chemicals; machinery; processed foods.

Traditional low-throughput manufacturing

Textiles and apparel; wood and furniture; printing; leather goods.

Prediction: The service-sector legacy should be stronger when manufacturing expansion creates intensive organizational and technical demands.

Classification follows Chandler (1990).

▸ Pictures

National Industry Growth and Local Industrialization

Combine each county's **1880 industry mix** with subsequent **national industry growth**:

$$Z_c^g = \sum_{j \in g} \underbrace{\omega_{cj,1880}^g}_{\text{1880 industry share}} \underbrace{G_{j,-c,1880\text{–}1900}}_{\text{leave-county-out growth}}, \qquad g \in \{A, H, L\}.$$

- **Three instruments:** all manufacturing (A), high throughput (H), and traditional low throughput (L).
- **Treatment and outcome:** instrument the manufacturing peak to explain HSPS specialization within services in 2000.
- **Conditioning:** state and peak-cohort fixed effects, 1850 characteristics, and complete 1850–1870 histories of HSPS employment shares.

Identifying assumption: Conditional on historical characteristics and pre-1880 trends, exposure to national industry growth affects local HSPS only through the industrialization it induces.

National Industry Growth and Persistent HSPS Services

Dependent variable: HSPS share of private-service employment in 2000.

	OLS	All industries IV	High-throughput IV	Low-throughput IV
Manufacturing peak	0.290**	1.330	2.830**	−10.950
	(0.120)	(1.050)	(1.180)	(80.100)
First stage, per 1 SD of shifter		1.50***	1.93***	0.06
		(0.39)	(0.54)	(0.42)
First-stage Conley F		14.67	12.87	0.02
Kleibergen–Paap F		20.53	21.94	0.03
Anderson–Rubin p-value		0.203	0.009	0.596
Counties	1,487	1,487	1,487	1,487

State/peak-cohort FE, 1850 controls and 1850–1870 histories. Conley SE (100 km); AR: Anderson–Rubin. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Census and QCEW.

▸ Rotemberg weights ▸ Measurement ▸ Shock inference ▸ Other checks ▸ Baseline checks ▸ Migration checks ▸ Event study

Model

Model Overview

- **Households:**
  - Non-homothetic consumption (Comin et al., 2020); fertility and education (Galor and Weil, 2000); supply skilled/unskilled labor.
- **Growth:**
  - Semi-endogenous during development (Jones, 1995): $g_j = (SL)^{\phi_j}$; endogenous in the long run: $g^{\infty} = \mathcal{G}(S^{\infty})$.
- **Production (New):**
  - Sectors $j \in \{a, m, c\}$; productivity $a \sim G(\mu, \sigma)$; setup cost F_j; exogenous exit (Hopenhayn, 1992). Inputs: intermediate goods, producer services.
  - Organization: Tasks $u \in [0, 1]$: **ad-hoc** (unskilled) vs. **formal** (skilled).

$$\underbrace{n \uparrow \Rightarrow \text{Formalization} \uparrow}_{\text{Coordination savings}} \Rightarrow K_{t+1} \uparrow \Rightarrow \underbrace{\text{Adaptation cost} \downarrow}_{\text{Externality to other firms}} \Rightarrow S^{\infty} \uparrow$$

Production

Gross output

$$y_{fjt} = Z_{jt} v_{fjt}^{\theta_{jt}} (m^a_{fjt})^{\gamma^a_{jt}} (m^m_{fjt})^{\gamma^m_{jt}} \left[P^p_{fjt}(q)\right]^{\gamma^P_{jt}}.$$

m^a, m^m: intermediate goods; $P^p(q)$: CES of skilled/unskilled producer services, with weights depending on q.

Value added

$$v_{fjt} = A_{jt} a_f \ell_{fjt}^{\eta}, \quad 0 < \eta < 1, \qquad n_{fjt} = \nu_{jt} \ell_{fjt}.$$

ℓ: CES composite of skilled/unskilled labor; n: employment (firm scale).

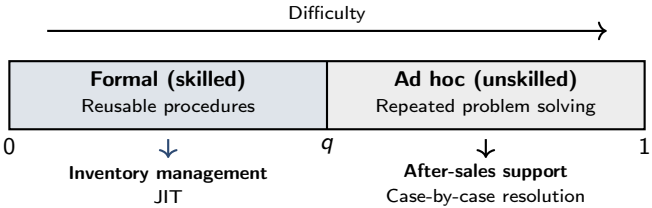
Recurring organizational costs (new)

$$C^o_t(n,q) = \underbrace{\frac{n}{\bar{N}}}_{\text{Task frequency}} \left[\underbrace{q p^H_t}_{\text{Formal}} + \underbrace{r(1-q) p^L_t}_{\text{Ad-hoc}} \right].$$

$q \in [0,1]$: share of formalized tasks; $r > 1$: repeated ad-hoc problem solving.

Choosing How to Organize Production

- Tasks $u \in [0, 1]$ ordered by difficulty; firms choose formalization cutoff q.



- Formalization: Upfront investment in reusable procedures (creation or adaptation).
- Recurring tasks: $n/\bar{N}$ instances per task; formal cost p_t^H vs. ad-hoc cost rp_t^L, $r > 1$.

Dynamic Externality: Creating and Adapting

- $K_t \in [0,1]$: share of tasks with publicly available organizational procedures.

Create a new procedure

$$C_t^C(z) = \frac{p_t^H}{\bar{e}_t^o} e^{z-1}$$

Necessary if $F(z) > K_t$.

Adapt an existing procedure

$$C_{jt}^A(z) = \frac{\rho}{\xi_j} C_t^C(z)$$

Available if $F(z) \leq K_t$. $\rho < 1$:

Knowledge diffusion

$$K_{t+1} = K_t + \delta_K \mathcal{E}_t$$

$\mathcal{E}_t$: employment-weighted exposure to newly created procedures.

Firm Scale and the Return to Formalization

Firm's investment decision

- Produce with inherited frontier s; invest to reach $u \geq s$ next period.

$$W_{jt}(a,s) = \pi^*_{jt}(a,s) + \max_{u \geq s} \left\{ - \underbrace{C^O_{jt}(s,u;K_t)}_{\text{One-time investment}} + \beta(1-\delta_f) W_{j,t+1}(a,u) \right\}.$$

Proposition 1: Scale–formalization complementarity

$$\frac{\partial \pi^*_{jt}(a,q)}{\partial q} = \underbrace{n^*_{jt}(a,q)}_{\text{Firm scale}} \quad \underbrace{\Psi_{jt}(q)}_{\text{Return per worker}}$$

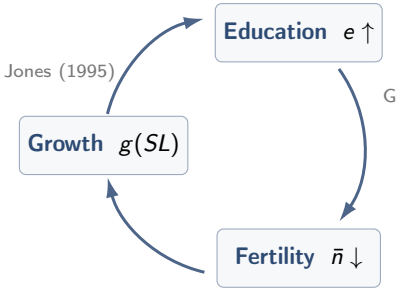
▸ Details ▸ Figure ▸ Proposition 1

Knowledge, Skills and Long-Run Growth

Organizational knowledge and skills

$$\text{Scale} \uparrow \implies K_t \uparrow \implies S^{\infty} \uparrow$$

Proposition 2: Demographic feedback and growth



$$\implies$$

$$g^{\infty} = \mathcal{G}(S^{\infty})$$

$$\frac{\partial \mathcal{G}}{\partial S} > 0$$

$$\bar{n}^{\infty} = 1 \qquad L^{\infty} < \infty$$

▸ HH Problem

Calibration

Calibration: The U.S. Development Path

1850–2019; annual model. Joint SMM: 19 parameters, 100 moments in 13 blocks.

Parameter block	Principal empirical discipline
Demand and productivity	Sectoral employment, manufacturing peak, GDP and VA productivity growth
Fertility and skills	Fertility, college attainment, earnings premium and VA skill shares
Organization: R, ξ_a, ξ_c	Sectoral HSPS employment shares
Scale: F_a, F_m, F_c	Employees per employer firm

Taking HSPS to the data: IPUMS + BEA Use Table

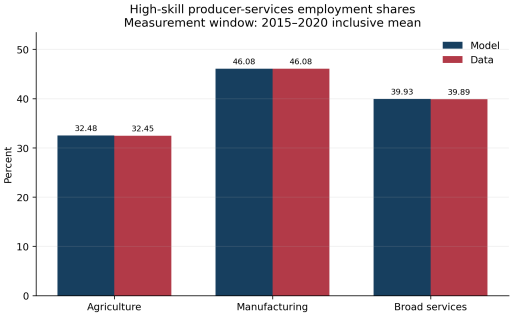
- Organizational and technical occupations; high skill: bachelor's degree or higher.
- Intermediate-service workers allocated to buyer sectors using BEA expenditure shares; final-consumer sales assigned to consumer services.

▸ Measurement ▸ Parameters ▸ Targeted fit ▸ U.S. paths

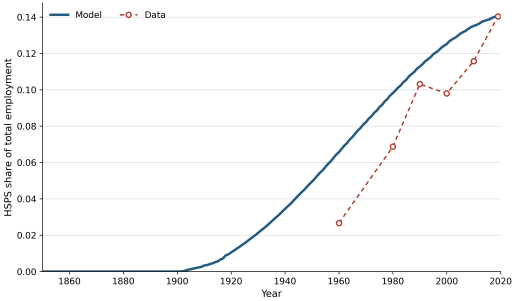
Fit and Validation

HSPS: Modern Fit and Historical Validation

Modern fit



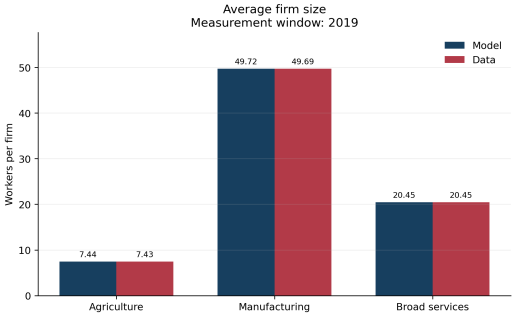
Historical validation



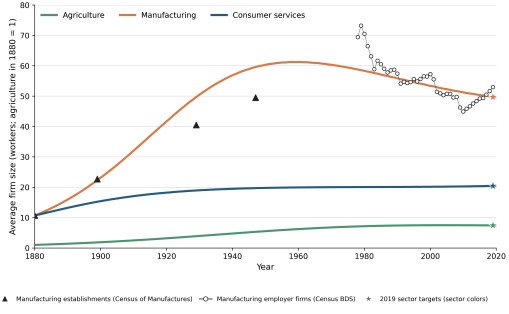
Left: targeted sectoral HSPS shares, 2015–2020 averages. Right: historical HSPS share of total employment. Source: IPUMS and BEA.

Firm Size: Modern Fit and Historical Validation

Modern fit

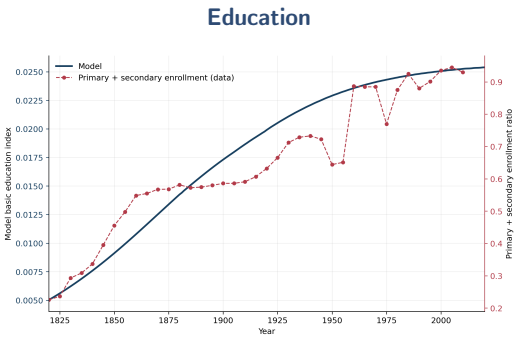


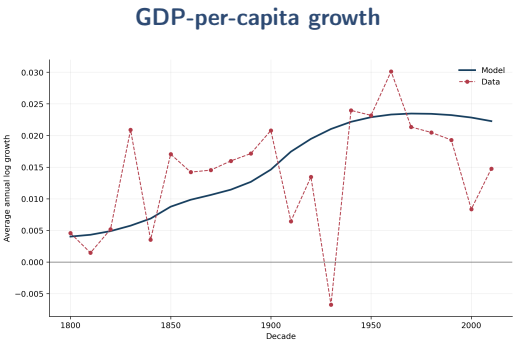
Historical validation



Left: targeted employees per employer firm, 2019. Right: historical manufacturing establishment and employer-firm sizes; initial scale anchored in 1880. Source: U.S. Census Bureau.

Untargeted Validation: Education and Growth





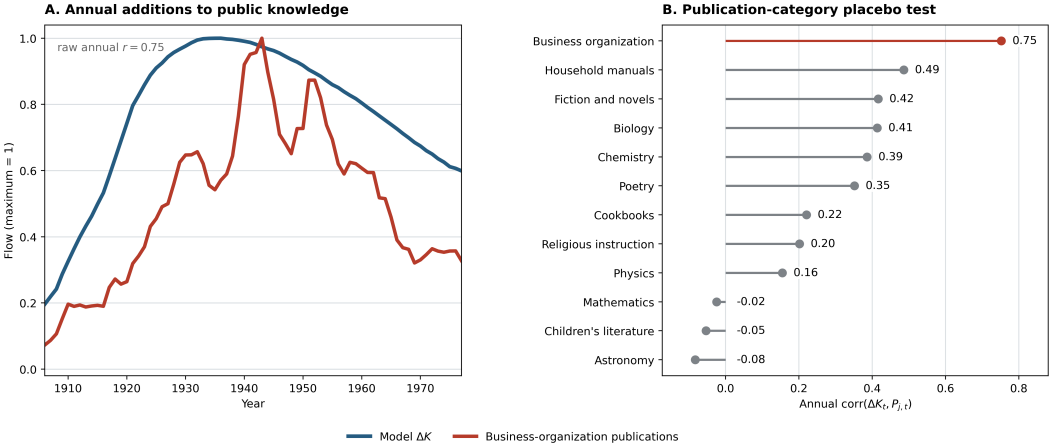
Left: model education index and primary–secondary enrollment. Right: annualized GDP-per-capita growth by decade, with the model shown as a smoothed trend. Neither series is directly targeted; GDP levels enter the calibration.

Measuring Public Organizational Knowledge

- Extract publications from the Library of Congress *Catalog of Copyright Entries*, 1906–1977.
- Count records containing fixed phrases/words per million OCR words. Compare to ΔK_t for various knowledge categories.

Publication category	**Examples of matched words and phrases**
Business organization	business systems; office management; personnel management; scientific management; time study; record-keeping...
Household manuals	household management; home economics; domestic science; housekeeping; household hints...
Chemistry	chemistry; chemical science; chemical analysis; biochemistry; electrochemistry...

Untargeted Validation: Public Organizational Knowledge



Copyright records per OCR word, 1906–1977; explicit manufacturing titles excluded. Lines: normalized five-year averages; correlations: annual data. Source: Library of Congress.

Quantitative Results

Counterfactuals: Manufacturing Histories and Development

1. **Vary manufacturing demand**
   Change only w_m throughout development; common initial conditions and all other parameters fixed.

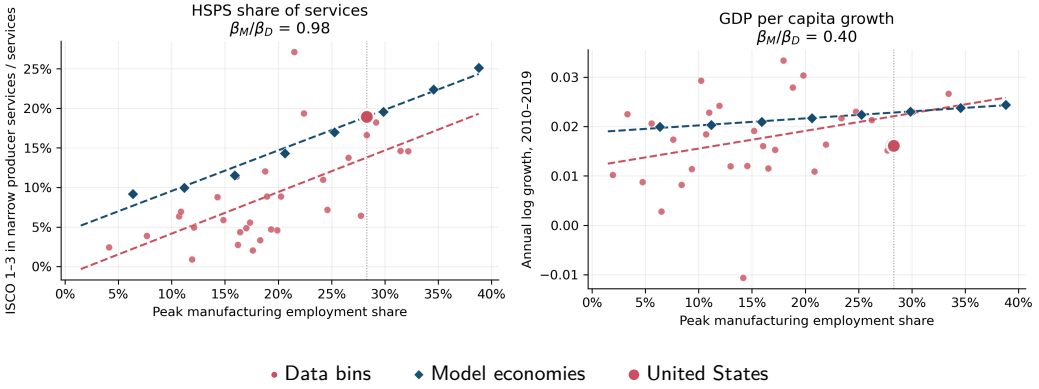
2. **Solve each development path**
   Manufacturing peaks, organizational knowledge, skills and income emerge in equilibrium.

3. **Compare model and data**
   Compare slopes relating each outcome to the historical manufacturing peak.

Outcomes: HSPS, tertiary attainment, income, GDP-per-capita growth and population growth.

▸ Demand-equivalent wedges

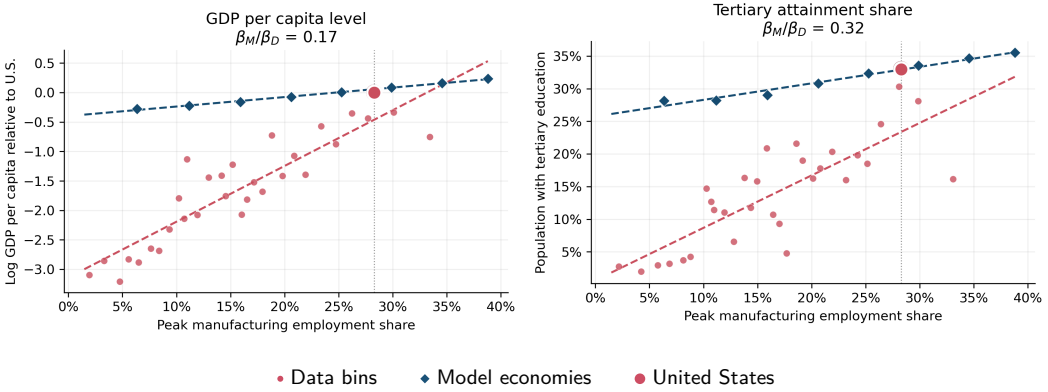
Manufacturing History, HSPS and Service-Era Growth



Data: 30 country bins; model: eight economies. Dashed lines: OLS on underlying observations; β_M/β_D: slope ratio. Samples: 48 countries (HSPS), 172 (GDP).

▸ HSPS measurement ▸ Sensitivity

Manufacturing History, Income and Skills



Data: 30 country bins; model: eight economies. Income: 2019; tertiary attainment: 2010 and 2015 average. Samples: 172 countries (GDP), 137 (education). Sources: Gapminder and Barro–Lee.

▸ Sensitivity ◂ Mechanism accounting ◂ Growth persistence

Industrial Policy

Policy: Temporary Support and Barriers to Firm Growth

Start from the same inherited LMIC state in 2026.

Temporary support

1. Skill scholarships
Subsidize the acquisition of skills.

2. Mixed package
Combine skill scholarships with support for organizational formalization.

Firm-scale environment

1. Distortions remain
Barriers to firm growth persist.

2. No distortions
Remove barriers to firm growth.

2026–2045
Temporary support

→

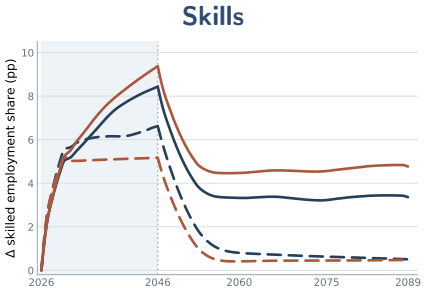
2046
Unexpected withdrawal

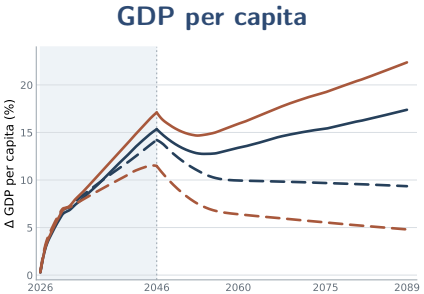
→

2089
Long-run legacy

▸ Model extension

Policy: What Remains After Withdrawal?





Policy: Skill scholarship — Mixed package **Environment:** No distortions - - - Distortions remain

Policy implication: Persistent gains require skills and an environment that supports firm growth.

Conclusion

Conclusion

1. **Industrial history shapes the service economy.**

   Larger manufacturing peaks predict more HSPS and faster subsequent GDP-per-capita growth.

2. **Production at scale builds transferable knowledge.**

   Firms develop reusable procedures that diffuse into services and sustain demand for skills.

3. **Scale reform and skills policies are complements.**

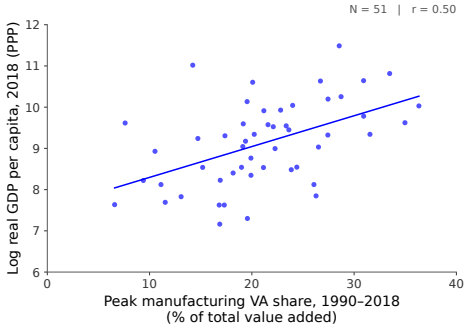
   Temporary support for skill acquisition and formalization has larger persistent effects when firms can grow.

Development without industrialization is possible,

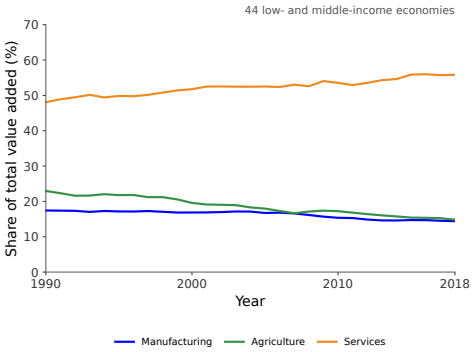
but the path into services matters.

Appendix

Motivation: Value Added



(a) Manufacturing VA peaks and income



(b) Sectoral value-added shares

Current-price VA shares, 1990–2018. Left: 51 economies; 2018 PPP income. Right: 44 low/middle-income economies, classified in 2018; unweighted means. Source: ETD and Gapminder.

Back

Household Problem: Fertility and Education

Preferences and child human capital

$$\max_{\{c_{ij,t}\}_{j\in\mathcal{J}},\, n_{i,t},\, e_{i,t}} \quad (1-\gamma)\log C_{i,t} + \gamma \log\left(n_{i,t} h_{i,t}\right)$$

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t} \qquad \sum_{j\in\mathcal{J}} \omega_j^{1/\sigma} \left(\frac{c_{ij,t}}{C_{i,t}^{\varepsilon_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1$$

Budget constraint

$$\sum_{j\in\mathcal{J}} p_{j,t}^{c} c_{ij,t} \leq \left[1 - n_{i,t}\left(\tau_i + \tau_e(S_t) e_{i,t}\right)\right] \bar{w}_t + d_t$$

$$\tau_e(S_t) = 1 - S_t(1-\tau_{es}), \qquad \frac{\partial \tau_e(S_t)}{\partial S_t} < 0$$

Back to Growth

Manufacturing Peaks and Growth in Service Economies

Dependent variable: average annual log GDP-per-capita growth, 2010–2019 (pp).

	All countries		Services > 50%		Peak before 2000 and services > 50%
	(1)	(2)	(3)	(4)	(5)
Manufacturing peak (10 pp)	0.629**	0.398	0.645*	1.113**	0.942*
	(0.263)	(0.459)	(0.345)	(0.515)	(0.490)
Mean service share (10 pp)	−0.572***	−0.540**	−1.045**	−1.313**	−1.450**
	(0.206)	(0.215)	(0.438)	(0.605)	(0.555)
Mean manufacturing share (10 pp)		0.354		−0.924	−1.199
		(0.536)		(0.883)	(0.730)
Region and income-group FE	Yes	Yes	Yes	Yes	Yes
Institutional quality control	Yes	Yes	Yes	Yes	Yes
Observations	173	173	104	104	90
R^2	0.360	0.362	0.278	0.288	0.372

HC3-robust standard errors in parentheses. Employment shares are averaged over 2010–2019. Column (5) additionally restricts recorded peaks to years before 2000. Income groups: 2009 GNI. Institutions: control-of-corruption percentile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: GGDC, ILOSTAT and World Bank.

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Manufacturing Peaks, Human Capital, and Service Composition

Dependent variable: average annual log GDP-per-capita growth, 2010–2019 (pp).

	Baseline (1)	HCQ (2)	HSPS (3)	Both (4)
Manufacturing peak (10 pp)	0.582*	0.264	0.340	0.167
	(0.343)	(0.315)	(0.303)	(0.305)
HCQ, 2015		0.841**		0.658**
		(0.351)		(0.325)
HSPS, 2018 (10 pp)			0.907*	0.624
			(0.493)	(0.455)
Mean service-employment share	Yes	Yes	Yes	Yes
Region and income-group FE	Yes	Yes	Yes	Yes
Observations	79	79	79	79
R^2	0.304	0.365	0.353	0.385
Peak-coefficient attenuation	—	54.7%	41.5%	71.4%

HC3 standard errors in parentheses; common sample of 79 countries. Service share: 2010–2019 average. Income groups: 2009 GNI. HSPS: ILO finance, business services and real estate / services excluding public administration, education and health. Attenuation relative to (1); joint HCQ–composition test in (4): $p = 0.028$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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County HSPS Measures and Employment Denominators

Inside manufacturing: occupations

Accounting and record keeping; purchasing, logistics and coordination; personnel and labor relations; technical and scientific work.

In services: industries

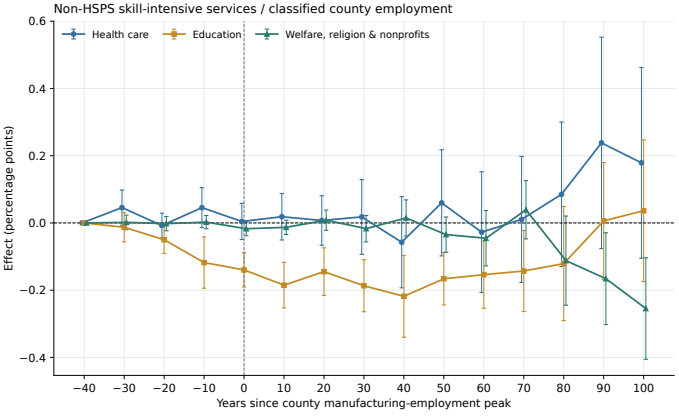
Information (NAICS 51); professional, scientific and technical services (54); management of companies and enterprises (55). Historical Census and SIC data use functional concordances.

Exercise	Denominator
Main event study	Classified employment: primary, manufacturing and services, reconstructed from the numerator's source in each vintage.
2000 cross section	QCEW private-service employment, excluding utilities.

Sources: Census of Manufactures, Population Census, CBP/EFSY and QCEW. Fixed 1850 county geography.

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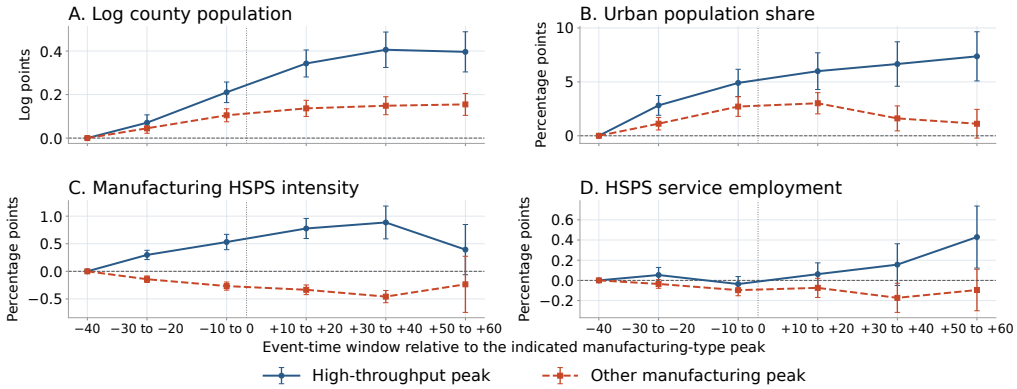
Falsification: Other HS Services



Service employment HS industries / classified employment. Same sample and specification as the HSPS-service panel; 95% CIs (Conley, 100 km). Source: Census and CBP.

◂ HSPS event study

Manufacturing Type: Development and HSPS Employment



Per 10 pp larger type-specific peak; 95% CIs (Conley, 100 km). HSPS denominators: manufacturing employment (C); classified private employment (D). Source: Census and CBP.

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Disclosure Treatment and Labor-Market Geography

High-throughput IV estimates under alternative outcome constructions.

Outcome and geography	N	Estimate	SE	First-stage F	AR p
QCEW public, counties, 2000	1,487	0.283**	0.118	12.87	0.009
QCEW flag-aware, counties, 2001	1,511	0.187*	0.100	11.25	0.066
EFSY native, counties, 2000	1,487	0.103	0.080	12.87	0.202
QCEW flag-aware, CZs, 2001	391	0.324**	0.141	9.03	0.005
EFSY harmonized, CZs, 2001	391	0.204	0.125	9.03	0.094
EFSY harmonized, CZs, 2010	391	0.279**	0.135	9.03	0.009

Preferred controls; Conley SE (100 km). EFSY: imputed CBP employment. CZ: commuting zone. AR: Anderson–Rubin. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Scale, Manufacturing HSPS, and Later Service Composition

Dependent variable: HSPS share of private-service employment in 2000.

	Baseline	+ Scale	+ Mfg. HSPS	+ Both
Manufacturing peak	0.480***	0.240****	0.300***	0.150
	(0.100)	(0.120)	(0.110)	(0.120)
Log workers per manufacturing establishment		0.600***		0.410*
		(0.220)		(0.230)
Manufacturing HSPS / classified employment			2.112***	1.927***
			(0.529)	(0.535)
Attenuation of peak coefficient (%)	0.0	50.3	37.4	68.3

1,487 counties; state/peak-cohort FE and 1850 controls. Conley SE (100 km). $^{*}p < 0.10$, $^{****}p < 0.05$, $^{***}p < 0.01$. Source: Census and QCEW.

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Inference with Common Industry Shocks

Outcome and inference	Estimate	SE	p-value	First-stage F	AR p
High-throughput instrument					
Public QCEW, 2000: county Conley	0.283	0.118	0.016	12.87	0.009
Public QCEW, 2000: exact leave-out	0.283	0.159	0.088	5.26	0.062
Public QCEW, 2000: common shocks	0.284	0.132	0.041	3.94	0.047
Flag-aware QCEW, 2001: exact leave-out	0.187	0.121	0.135	5.02	0.065
Flag-aware QCEW, 2001–03: exact leave-out	0.164	0.108	0.142	5.39	0.082
All-industry instrument					
Public QCEW, 2000: exact leave-out	0.133	0.223	0.551	1.59	0.623

Exact leave-out: industry-clustered scores. Common shocks: leave-sample-out growth. 26 high-throughput industries; effective count: 4.71. Tests use finite-industry degrees of freedom. AR: Anderson–Rubin.

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Classification, Exposure, and Conditioning Checks

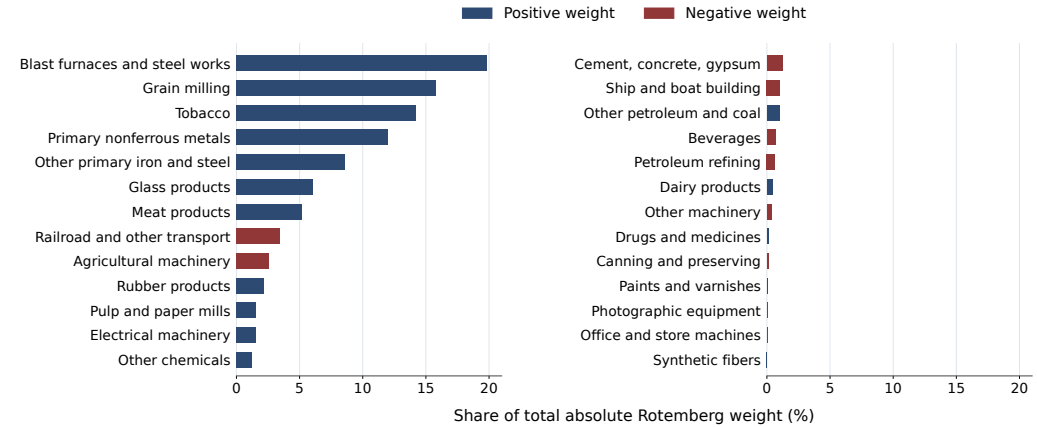
High-throughput IV; QCEW public-record outcome in 2000.

Specification	Estimate	SE	p-value	First-stage F	AR p
Exclude processed foods	0.325	0.257	0.207	3.63	0.159
Exclude machinery	0.189	0.098	0.055	28.98	0.041
Narrow continuous/process set	0.183	0.078	0.018	29.69	0.011
Add historical geography controls	0.320	0.120	0.008	13.43	0.002
Add manufacturing employment share in 2000	0.196	0.096	0.042	14.81	0.042
Preferred controls; Conley radius 500 km	0.283	0.170	0.096	2.83	0.067

Preferred controls except where indicated. Conley SE, 100 km unless stated otherwise. AR: Anderson–Rubin.

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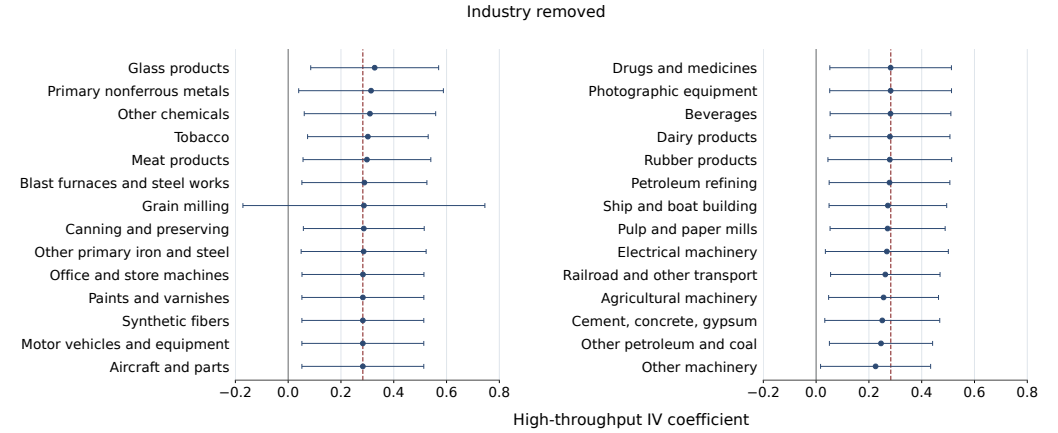
Which Industries Identify the High-Throughput IV?



26 industries; absolute Rotemberg weights after residualizing on the preferred controls. The five largest account for 70.4%; negative weights account for 10.2% of total absolute weight. Source: Census and QCEW.

Leave one out Back

High-Throughput IV: Leave One Industry Out



Each row drops the named industry and rebuilds the shifter. 2SLS estimates with 95% Conley intervals (100 km); preferred controls. The dashed line is the benchmark. Source: Census and QCEW.

◂ Weights ◂ IV results

Historical Balance and IV Robustness

A. Pre-1880 HSPS growth: reduced form per one-SD shifter

Outcome and period	Estimate	SE	*p*-value
Manufacturing HSPS / classified employment, 1850–1860	0.0012	0.0007	0.100
Manufacturing HSPS / classified employment, 1860–1870	0.0006	0.0007	0.399
HSPS services / private-service employment, 1850–1860	0.151	0.171	0.378
HSPS services / private-service employment, 1860–1870	−0.225*	0.122	0.064

B. High-throughput IV: effect of a 10-pp larger manufacturing peak (pp)

Specification	Estimate	SE	First-stage *F*	AR *p*-value	*N*
Baseline	2.83**	1.18	12.87	0.009	1,487
Detailed historical geography	3.20***	1.20	13.43	0.002	1,487
Railroad access in 1880	3.02**	1.32	10.55	0.009	1,487
Exclude top-10 cities	2.70**	1.17	12.92	0.013	1,474
Commuting zones	3.24**	1.41	9.03	0.005	391

Notes: Panel A reports reduced forms per one-SD shifter, conditional on state fixed effects, 1850 controls, and other pre-1880 histories; $N = 1{,}487$. Panel B uses public QCEW (2000); the commuting-zone row uses flag-aware QCEW (2001) and reaggregated treatment and shifter. Geography adds coal, waterways, ports, fall-line, and Great Lakes access. Standard errors are 100-km Conley; AR denotes Anderson–Rubin. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

◂ IV results

Geographic Reach and Migrant Origins

A. Own-county and surrounding-area manufacturing peaks

	Own only	50 km	100 km	200 km	300 km	500 km
Own-county peak	0.29**	0.20*	0.26**	0.24**	0.26**	0.27**
	(0.12)	(0.12)	(0.12)	(0.12)	(0.11)	(0.12)
Area peak within radius		0.22	0.13	0.23	0.90*	0.63
		(0.20)	(0.22)	(0.24)	(0.50)	(0.65)

B. Industrialization exposure at migrants' counties of origin

	Baseline	+ Migrant-origin industrialization
Own-county peak	0.29**	0.16
	(0.12)	(0.12)
Average origin peak among migrants		1.00***
		(0.37)

Notes: The outcome is the public-QCEW HSPS service share in 2000; $N = 1{,}487$. Peak effects are percentage points per 10-percentage-point larger peak. Area peaks aggregate employment within each radius before taking the 1880–1940 maximum. Migrant exposure is the one-SD average origin-county peak among linked movers and is post-treatment. All regressions include baseline controls and fixed effects; standard errors are 100-km Conley. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

◂ IV results

Manufacturing Peaks and HSPS Employment

Dependent variable (pp)	HSPS / total employment Change, 1990–2018					HSPS / market services Level, 2018
	(1)	(2)	(3)	(4)	(5)	(6)
Manufacturing peak (10 pp)	2.867***	2.937***	2.173***	1.951*	1.945*	4.521***
	(0.507)	(0.501)	(0.782)	(1.041)	(1.058)	(1.273)
GDP-per-capita gain ($\Delta \ln y$)		−1.202	−0.176	−0.108	−0.013	
		(0.732)	(1.029)	(1.113)	(1.299)	
Log GDP per capita, 1990				0.531	0.771	
				(1.036)	(1.263)	
HSPS share, 1990 (pp)					−0.093	
					(0.327)	
Region and initial income-group FE	No	No	Yes	Yes	Yes	Yes
Economies	51	51	51	51	51	51
R^2	0.530	0.550	0.617	0.620	0.622	0.681

HSPS: business services, finance and real estate. Market services exclude government. The 1990 HSPS control uses total employment. Unweighted OLS; HC3 standard errors in parentheses. Income groups use 1990 GNI. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Sources: ETD, GGDC and Maddison (2023).

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Why Scale Raises the Return to Formalization

1. Operating profit

After optimizing over intermediate inputs and producer services:

$$\pi^*(a,q) = \max_{\ell \geq 0} \left\{ p^v(q) A a \ell^\eta - [w^v + c^o(q)]\ell \right\}, \qquad n = \nu \ell,$$

where

$$c^o(q) = \frac{\nu}{\bar{N}} \left[q p^H + r(1-q) p^L \right].$$

2. Envelope theorem

$$\frac{\partial \pi^*}{\partial q} = \underbrace{v^* \frac{\partial p^v(q)}{\partial q}}_{\substack{\text{Productive-service} \\ \text{effect}}} + \underbrace{\frac{n^*}{\bar{N}} (r p^L - p^H)}_{\substack{\text{Recurring} \\ \text{cost savings}}}.$$

◂ Scale result

Organizational Investment, Diffusion and Entry

$$\mathcal{C}^{O}_{jt}(s,u;K_t)=\int_s^u \widetilde{C}_{jt}(z;K_t)\,d\mathcal{F}(z),\qquad \widetilde{C}_{jt}(z;K_t)=\begin{cases}\min\{C^{C}_{t}(z),C^{A}_{jt}(z)\}, & \mathcal{F}(z)\le K_t,\\ C^{C}_{t}(z), & \mathcal{F}(z)>K_t.\end{cases}$$

$$\mathcal{E}_t=\frac{1}{N_t}\sum_j\int n^{*}_{jt}(a,s)\left[\int_{\mathcal{F}(z)>K_t} c_{jt}(a,s;z)\,d\mathcal{F}(z)\right]d\mu_{jt}(a,s).$$

$c_{jt}(a,s;z)$ indicates a privately created procedure still held by the firm; exposure counts only procedures not yet public.

$$N_t=\sum_j\int n^{*}_{jt}(a,s)\,d\mu_{jt}(a,s),\qquad K_{t+1}=\min\{1,K_t+\delta_K\mathcal{E}_t\}.$$

$$I_{jt}\ge 0,\quad F_j I_{jt}^{\phi_E}-V^{E}_{jt}\ge 0,\quad I_{jt}\left[F_j I_{jt}^{\phi_E}-V^{E}_{jt}\right]=0.$$

$$V^{E}_{jt}=\int W_{jt}(a,0)\,dG(a),\qquad M_{jt}=(1-\delta_f)M_{j,t-1}+I_{jt},\qquad \overline{n}_{jt}=\frac{N_{jt}}{M_{jt}}.$$

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Aggregate Dynamics: Annual Implementation

Cohort replacement

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \quad \mathcal{S}' > 0, \qquad S_{t+1} = \left(1 - \frac{1}{\Delta}\right) S_t + \frac{1}{\Delta} p_t^S.$$

Population

$$L_{t+1} = L_t \bar{n}_t^{1/\Delta}, \qquad \bar{n}_t = \sum_i \lambda_{i,t} n_{i,t}.$$

Sectoral and aggregate productivity growth

$$g_{j,t}^A = \left[1 + \lambda_A (S_t L_t)^{\phi_j}\right]^{1/\Delta} - 1, \qquad g_t = \sum_{i \in \{a,m,s\}} \lambda_{i,t} g_{i,t}^A.$$

Δ: generation length; $\lambda_{i,t}$: employment shares; Π_{t+1}: anticipated skilled-wage premium. Consumer and producer services share ϕ_c; $0 < \phi_j < 1$.

Households and skill choice | Equilibrium | Dynamics

Household Decision Rules and Skill Acquisition

Fertility and continuous education

$$e^*_{i,t} = \left[\sqrt{g_t\left(\frac{\tau_i}{\tau_e(S_t)} - \tau_h\right)} - \tau_h\right]_+, \qquad n_{i,t} = \frac{\gamma\zeta_t}{\tau_i + \tau_e(S_t)e^*_{i,t}}, \qquad \zeta_t = 1 + \frac{d_t}{\bar{w}_t}.$$

$$\bar{n}_t = \sum_i \lambda_{i,t} n_{i,t}, \qquad \tau_e(S_t) = 1 - S_t(1 - \tau^s_e).$$

Discrete skill choice of entering cohorts

$$p^S_t = \left[1 + \exp\left(-\frac{\beta_\pi \log \Pi_{t+1} - c^{\text{skill}}}{\sigma_{\text{skill}}}\right)\right]^{-1}.$$

c^{skill}: skill-acquisition cost; σ_{skill}: dispersion of idiosyncratic choice costs.

$[x]_+ = \max\{x, 0\}$. Education rule shown under $\tau_i/\tau_e(S_t) > \tau_h$; firms' organizational education input uses education chosen 30 years earlier.

◂ Model overview ◂ Dynamics

Perfect-Foresight Competitive Equilibrium

Given initial population, skills, knowledge, productivities and firm distributions:

1. Households choose consumption, fertility, education and cohort skill.
2. Firms choose inputs, employment and next-period formalization.
3. Entry satisfies free entry; producer-service suppliers are competitive.
4. Final goods, both producer services and both labor markets clear.
5. Aggregate states and firm distributions follow their laws of motion.
6. Anticipated aggregate prices and states equal their realizations.

$$\mu_{j,t+1}(B) = (1-\delta_f)\int 1\{(a, u^*_{jt}(a,s)) \in B\}\, d\mu_{jt}(a,s) + I_{j,t+1}\int 1\{(a,0)\in B\}\, dG(a).$$

Manufacturing is the numeraire. Idiosyncratic ability and exit risk; no aggregate uncertainty. Transition solved by backward firm optimization and forward aggregation.

◂ Dynamics

Population and Long-Run Growth

For a terminal skilled share S, fertility converges to replacement: $\bar{n}^{\infty} = 1$.

$$g^{\infty} = \mathcal{G}(S;\zeta) = \underbrace{\frac{(\gamma\zeta - \tau_s)^2}{\tau_s}}_{\text{Demographics}} \times \underbrace{\frac{1}{\tau_e(S)}}_{\text{Education efficiency}}.$$

Population adjusts to a unique finite level:

$$L^{\infty}(S;\zeta) = \frac{1}{S}\left[\frac{[1+\mathcal{G}(S;\zeta)]^{\Delta} - 1}{\lambda_A}\right]^{1/\phi_c}.$$

$$\tau_e(S) = 1 - S(1-\tau_e^s), \qquad 0 < \tau_e^s < 1 \quad \Longrightarrow \quad \frac{\partial \mathcal{G}(S;\zeta)}{\partial S} > 0.$$

Terminal service economy; $\tau_h = 0$, $\gamma\zeta > \tau_s$, and fixed full-income ratio $\zeta = 1 + d^{\infty}/\bar{w}^{\infty}$. Δ: generation length.

▸ Proof ▸ Manufacturing legacy ◂ Dynamics

Proposition 2: Replacement Fertility and Growth

Fix S and ζ in the terminal service economy; set $\tau_h = 0$.

$$g(L) = \left[1 + \lambda_A (SL)^{\phi_c}\right]^{1/\Delta} - 1, \quad e(L) = \sqrt{\frac{\tau_s g(L)}{\tau_e(S)}}, \quad n(L) = \frac{\gamma\zeta}{\tau_s + \tau_e(S) e(L)}.$$

$g(L)$ is increasing and $n(L)$ decreasing, with $n(0) > 1$ and $n(\infty) = 0$.

$$T(L) = L n(L)^{1/\Delta}, \qquad 0 < 1 - \frac{\phi_c}{2\Delta} < \frac{d \log T(L)}{d \log L} < 1.$$

At the unique fixed point, $n = 1$:

$$e^{\infty} = \frac{\gamma\zeta - \tau_s}{\tau_e(S)}, \qquad g^{\infty} = \frac{(\gamma\zeta - \tau_s)^2}{\tau_s \tau_e(S)}, \qquad \frac{\partial g^{\infty}}{\partial S} > 0.$$

$\gamma\zeta > \tau_s$, $0 < \phi_c < 1$ and $\Delta \geq 1$. The population map is increasing and moves toward the fixed point from either side.

◂ Proposition 2

Manufacturing and Long-Run Growth

Economy A Agriculture $\longrightarrow$ Manufacturing $\longrightarrow$ Services

Economy B Agriculture $\longrightarrow$ Services

$$g_A^{\infty} > g_B^{\infty} \iff S_A^{\infty} > S_B^{\infty} \iff K_A^{\infty} > K_B^{\infty}$$

$$\iff \underbrace{\frac{\tau_m}{\tau_s}}_{\text{education}} \underbrace{(S_1 L_1)^{\phi_m - \phi_c}}_{\text{productivity growth}} \underbrace{\left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}}}_{\text{firm scale}} > 1.$$

Common initial state; three-stage comparison with symmetric firms, common prices and terminal full-income ratio, and nonsaturated service formalization; $\tau_h = 0$, $\Delta = 1$.

▸ Proof and conditions ◂ Long-run growth

Proposition 3: The Manufacturing Legacy

Under the three-stage comparison:

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad \bar{e}_m^o n_m > \bar{e}_c^o n_c.$$

$$\frac{\bar{e}_m^o}{\bar{e}_c^o} = \left[\frac{\tau_m}{\tau_s}(S_1 L_1)^{\phi_m - \phi_c}\right]^{1/2}, \qquad \frac{n_m}{n_c} = \left(\frac{F_m}{F_c}\right)^{\frac{1}{1+\phi_E}}.$$

In the terminal service economy, additional public procedures raise formalization on an unsaturated margin:

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad S_A^\infty > S_B^\infty \quad \Longleftrightarrow \quad g_A^\infty > g_B^\infty.$$

Common initial states, prices, operating-surplus terms and terminal full-income ratio; symmetric static firms; $\tau_h = 0$, $\Delta = 1$. Nonsaturation requires that an additional public procedure changes formalization for some service firms.

◂ Proposition 3

HSPS: Measurement in Calibration and Cross-Country Data

U.S. calibration: use-based sectoral allocation

- In-house organizational and technical occupations, plus purchased producer services.
- Bachelor's degree or higher defines high skill.
- BEA intermediate-use expenditure shares allocate service suppliers to destination sectors; final sales enter consumer-service value added.

Cross-country accounting: harmonized industry–occupation proxy

$$\frac{\text{ISCO-08 occupations 1–3 in ISIC divisions 62–66, 69–75}}{\text{Employment in market services: ISIC sections G–N, R–S}}$$

ILOSTAT, 2010–2019 averages; 48 countries.

The U.S. allocation assumes common skill composition and labor requirements across an industry's buyers. Aggregate HSPS is an accounting aggregate of sectoral targets.

◂ Calibration ◂ Cross-country results

Calibration: Parameter Estimates and External Benchmarks

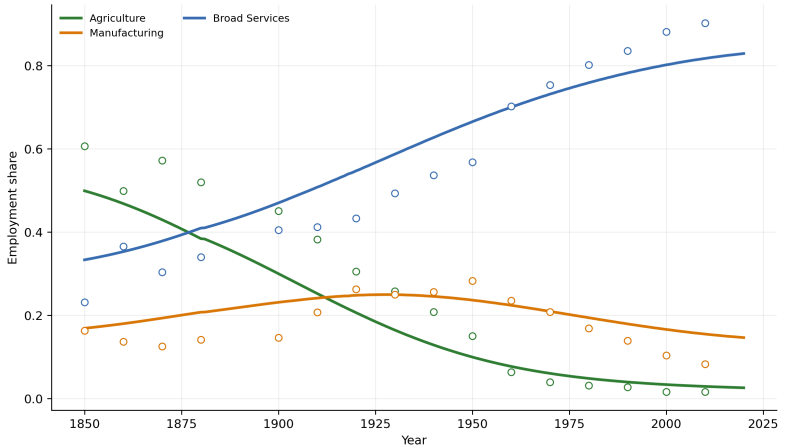
Block	Jointly estimated parameters	Estimates
Fertility	τ, τ_a^n	0.0324, 0.4402
Productivity	$\phi_a, \phi_m, \phi_c, g_{GO}$	0.9984, 0.9192, 0.6905, 0.00318
Skills	$c_{\text{col}}, \alpha_a, \alpha_m, \alpha_c$	0.5682, 0.2434, 0.3300, 0.4100
Demand	ϵ_c, w_a, w_c	1.0105, 0.6510, 3.8233
Organization	R, ξ_a, ξ_c	22.79, 3.5640, 6.8774
Entry	F_a, F_m, F_c	25.872, 187.625, 85.117

Selected assigned parameters	Value / empirical anchor
δ_K	0.09; diffusion benchmark (Mansfield, 1961)
ρ	0.40; maintained adaptation/creation cost ratio
$\bar{N}$	25; recurring coordination uses about 5% of employment
σ_{skill}	0.10; college attainment and earnings-premium changes
η; σ	0.85 (span of control); 0.30 (final-good substitution)

Joint SMM. The nonredundant Jacobian has rank 19; effective rank 17 at a 1% cutoff. Service adaptability is weakly pinned down within its saturation region.

◂ Calibration

Fit to U.S. Structural Change



Lines: model; circles: data. Employment paths are targeted. Manufacturing peak: 24.93% in 1928 (model), 28.28% in 1950 (data).

▸ Targeted fit ◂ Calibration

Calibration: Targeted Scalar Moments

Moment	Data	Model
HSPS share: agriculture	0.3245	0.3248
HSPS share: manufacturing	0.4608	0.4608
HSPS share: consumer services	0.3989	0.3993
Employees per firm: agriculture	7.435	7.441
Employees per firm: manufacturing	49.694	49.717
Employees per firm: services	20.454	20.454
VA productivity growth: agriculture	0.0371	0.0385
VA productivity growth: manufacturing	0.0317	0.0298
VA productivity growth: consumer services	0.0142	0.0131
Aggregate skilled share	0.3299	0.3247
Skilled earnings premium	1.600	1.645
Peak manufacturing-employment share	0.2828	0.2493
Agriculture/service fertility ratio	2.000	1.990

Shares and annual growth rates in fractions. HSPS: 2015–2020; firm size: 2019; productivity growth: 1970–2010. Employment, fertility and GDP paths also enter SMM.

◂ Calibration ◂ U.S. fit

Manufacturing Demand and Equivalent Consumption Wedges

At fixed income and producer prices, define

$$f_m \equiv \frac{w_m^{\mathrm{CF}}}{w_m^\star}, \qquad k_j \equiv \frac{\varepsilon_j}{\varepsilon_m}, \qquad j \in \{a, c\}.$$

Match the relative-demand terms

$$\underbrace{\left(\frac{p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{\left(f_m w_m^\star\right)^{k_j}}}_{\text{Demand shifter}} = \underbrace{\left(\frac{(1+\tau_j)p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{\left(w_m^\star\right)^{k_j}}}_{\text{Consumption wedge}}$$

$$(1+\tau_j)^{1-\sigma} = f_m^{-k_j} \quad \Longleftrightarrow \quad \boxed{1+\tau_j = f_m^{-k_j/(1-\sigma)}}.$$

◂ Experiment ◂ Growth paths

Sensitivity of the Cross-Country Gradients

18 economic variants: all 90 slopes retain their benchmark sign.

	GDP level	GDP growth	Tertiary	HSPS	Population growth
Median absolute change (%)	2.9	8.5	10.2	2.4	4.3
Diffusion grid: $\delta_K \in [0.04, 0.14]$					
Minimum change (%)	−13.8	−6.9	−27.8	−22.4	−16.2
Maximum change (%)	0.6	5.0	4.1	1.6	0.6

- Broad variants: adaptation costs, diffusion, scale, substitution, ability dispersion, education costs and knowledge accumulation.

◂ HSPS and growth ◂ Mechanism accounting

When can a temporary intervention change an economy's growth path?

Recalibrate to a median LMIC with small establishments and scarce skills.

Endogenous establishment setup

Before drawing ability a, entrants choose setup intensity h:

$$x = ha, \qquad h \in [\underline{h}, 1].$$

Lower h reduces productivity and the setup inputs needed to enter.

Distortions against productive firms

More productive establishments face larger revenue wedges:

$$1 - \tau_i = \kappa_j x_i^{-d_j}, \qquad d_j = 0.41.$$

Removing d_j restores $h = 1$ and the benchmark entry block.

◂ Policy experiment

Temporary Policies: Costs and Outcomes

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Distortions remain						
Skill acquisition scholarships	0.593	2.59	5.56	0.00	0.01	0.000
Formalization support	0.017*	0.18	2.13	0.27	0.13	0.036
Mixed package	0.613	6.95	8.02	0.00	0.02	0.000
HSPS wage subsidy	0.101†	−3.31	−1.51	0.10	0.04	0.009
Manufacturing wage subsidy	0.598	−3.48	−2.14	0.00	−0.01	0.000
Full permanent distortion reform						
Skill acquisition scholarships	0.581	6.92	8.52	1.64	0.29	0.026
Formalization support	0.123*	1.67	8.43	8.63	1.46	0.256
Mixed package	0.600	7.51	10.78	4.48	0.75	0.078
HSPS wage subsidy	0.580	0.57	1.79	3.15	0.55	0.052
Manufacturing wage subsidy	0.597	−0.16	−0.42	0.00	0.00	0.000

PV cost: discounted spending / discounted GDP in 2026–2045. CEV: 2026–2089; other outcomes: 2089, relative to each environment's no-policy path. *100% reimbursement cap. †Fiscal spending jumps from 0.101% to 2.083% across the threshold.

◂ Policy results

Full Reform with the U.S. Entry Block

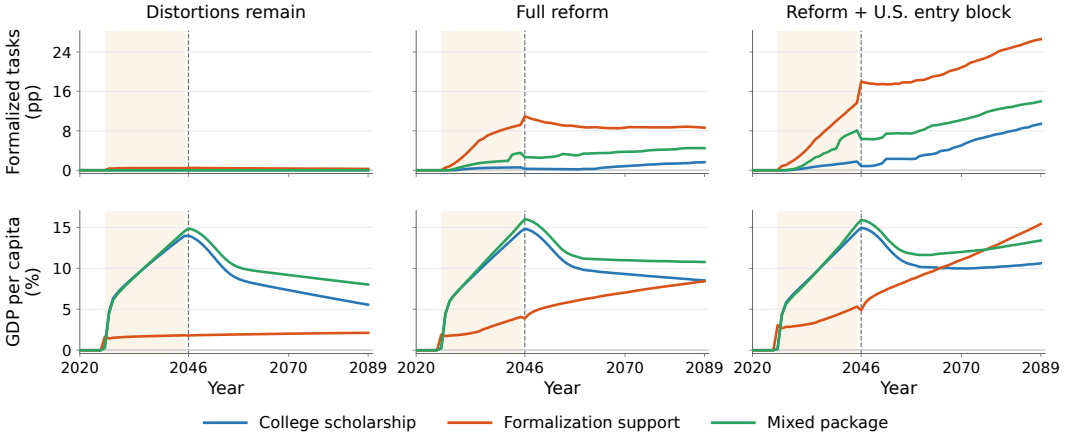
From the common inherited 2026 state: $F_m = 187.625$, $F_c = 85.117$.

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Skill acquisition scholarships	0.579	7.17	10.62	9.40	1.41	0.069
Formalization support	0.130*	2.30	15.43	26.60	4.04	0.381
Mixed package	0.585	7.77	13.40	13.98	2.10	0.140
HSPS wage subsidy	0.620	0.70	3.82	11.02	1.63	0.086
Manufacturing wage subsidy	0.597	−0.15	−0.41	0.00	0.00	0.000

Policies operate in 2026–2045. CEV: 2026–2089; other outcomes: 2089. Effects are relative to the no-policy path with full reform and U.S. entry costs. *Formalization support reaches the 100% reimbursement cap.

◂ Policy results

Policy Withdrawal: Formalization and Income



Changes relative to each environment's no-policy path, 2020–2089. Shading: temporary support in 2026–2045; dashed line: unexpected withdrawal in 2046.

Skills and knowledge

Manufacturing History and Subsequent Growth

High peak: 29.88% **Low peak: 11.19%**

		2030	2050	2070
Skilled-employment share (%)	High	34.18	34.64	35.02
	Low	28.74	29.37	29.83
GDP-per-capita growth (% p.a.)	High	2.24	2.22	2.20
	Low	1.99	2.01	2.03
GDP per capita (low peak = 100)	High	139.44	145.57	150.89

Only w_m differs: 3.49 (high), 0.60 (low). Growth is the annualized log change in chain-weighted real GDP per capita over the preceding decade.

◂ Cross-country accounting

Mechanism Accounting

Repeat the manufacturing-demand experiment, changing one mechanism at a time.

	Share of baseline slope (%)				
	HSPS share	Tertiary attainment	GDP per capita	GDPpc growth	Population growth
	2010–2019	2010–2019	2019	2010–2019	1850–2019
Baseline	100.0	100.0	100.0	100.0	100.0
Equal entry barriers, F_j	12.7	−38.4	49.0	40.7	36.0
No knowledge diffusion, $\delta_K = 0$	47.2	26.5	71.3	67.6	67.9
Equal productivity elasticities, ϕ_j	119.1	81.8	76.4	118.7	1.5
Equal adaptability, ξ_j	83.6	77.9	92.7	89.7	93.0
Equal VA skill intensity, α_j	99.7	113.0	108.8	99.8	122.8

Common five-economy support; slopes use each counterfactual's realized manufacturing peak. Other parameters fixed; no recalibration.

▸ Sensitivity ◂ Cross-country accounting

Service Industries in the ETD

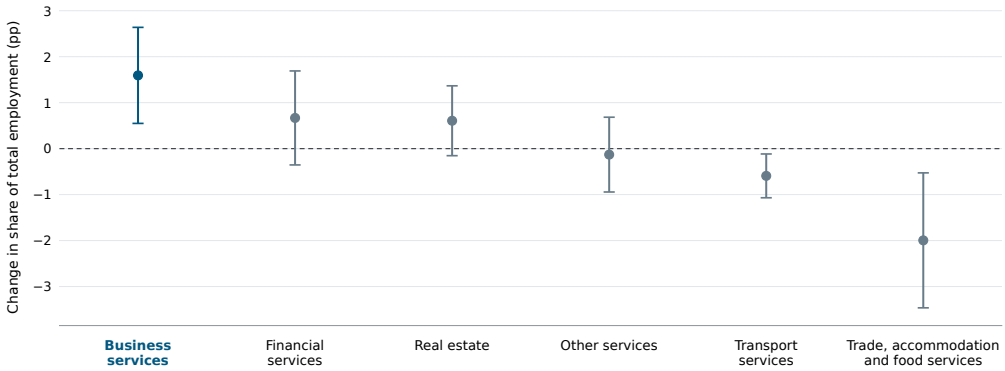
Figure category	ISIC Rev. 4	Activities included: examples
Business services	J, M, N	IT, telecoms and media; accounting, consulting, engineering and R&D; recruitment, equipment rental, cleaning and security.
Financial services	K	Banking, insurance, pension funds and related financial services.
Real estate	L	Property rental, brokerage and management.
Other services	R, S, T, U	Arts, recreation, personal services, household repairs and associations; domestic workers, household activities and extraterritorial bodies.
Transport services	H	Passenger and freight transport, warehousing, postal and courier services.
Trade, accommodation and food services	G, I	Wholesale and retail, motor-vehicle repair, hotels, restaurants and catering.

Industry totals include all occupations. The figure excludes O–Q: public administration, education, health and social work.
Source: GGDC/UNU-WIDER ETD; UN ISIC Rev. 4.

◂ Service-industry decomposition

Stylized Fact 1: Service-Industry Decomposition

Change in employment share, 1990–2018, per 10 pp larger manufacturing peak.



51 economies; separate OLS regressions, no additional controls. Shares of total employment; 95% CIs using HC3 standard errors. Source: ETD and GGDC.

▸ Industry definitions ▸ Back

What Does “High Throughput” Mean?

LOW THROUGHPUT

Textiles & apparel



Scale by adding producers

HIGH THROUGHPUT

Iron & steel



Scale by coordinating a system

Back

Firm Scale and Organizational Choice

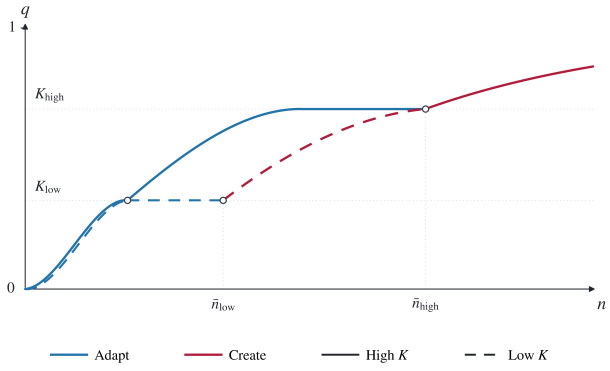


Figure 3: Blue is adaptation, red is creation of organizational knowledge. In a high knowledge economy K, firms formalize more activities drawing on a larger set of organizational methods they can adapt. Very large firms always create knowledge.

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